ADULT HYPERTHERMIA ALGORITHM

GUIDELINES 2025
EUROPEAN RESUSCITATION COUNCIL©

CPR

Require CPR — YES → Universal ALS algorithm

NO

Core temperature

Core temperature* > 40.5° C — YES → Rapid cooling (cold water immersion) → Stop cooling at core temperature <39°C → Continue monitoring for at least 15 min after cooling

- Rehydrate as required
- Check for improved mental status
- Avoid accidental hypothermia (< 35°C)

NO

Core temperature ≤ 40.5° C & confused / disoriented — YES → Rapid cooling (cold water immersion)

NO

Fluids level

Blood sodium < 130 mEq L^{-1} — YES → If abnormal mental state, initiate IV 100ml bolus of 3% saline at 10 min intervals, 2nd and 3rd bolus only if required. If normal mental state administer oral sodium → Hyponatraemia algorithm

NO

Severely dehydrated?** — YES → If abnormal mental state, administer IV normal saline or Ringer's lactated. If normal mental state provide oral rehydration and sodium

NO

Other symptoms

Other symptoms — YES → Appropriate algorithm e.g. hypoglycaemia

NO

Release with exercise restrictions*** to cool environment and oral fluids

TIME IS KEY: COOL AND RUN APPROACH

- Cool first, then transfer to hospital after
- Immediate cooling
- Rapid cool to < 39°C until symptoms resolve

Use a temperature probe

Bathtub ½ to ¾ filled with water & ice, 1-17° C, stirred or circulated

ALS – advanced life support; CPR – cardiopulmonary resuscitation; IV - intravenous; *Core temperature should be measured, e.g. tympanic, rectal, or oesophageal; ** Dry mouth, thirsty, hypotensive; ***Return to activity considerations following a diagnosis of exertional heat illness and exertional heat stroke have been published recently elsewhere